

Wood Products-II

□ To provide basic knowledge regarding the manufacturing process of Wood based

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5122 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5122.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	5122
Course title	Wood Products-II
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide basic knowledge regarding the manufacturing process of Wood based panel products such as Block board, particle board, fibre board, oriented strand
- To make aware of the details of their properties and testing.

Course outcomes

CO	Official outcome	Study level
CO1	15 Applying testing and properties. Disseminate pressing, finishing, testing and	See official syllabus
CO2	14 Applying properties of particle board Explain the Chipping of wood, pulping methods,	See official syllabus
CO3	Mat formation manufacture of Insulation board and 15 Applying Hardboard Explain sandwich composition, bulking treatments	See official syllabus
CO4	14 Applying and testing and properties of wood laminates	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 1 2
CO2	CO2 3 2
CO3	CO3 3
CO4	CO4 3 2

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	properties	4	As specified
Module 2	Disseminate pressing, finishing, testing and properties.	4	As specified
Module 3	manufacture of Insulation board and Hardboard	4	As specified
Module 4	properties of wood laminates.	3	As specified

MODULE 1

properties

This module is presented from the official syllabus scope for Wood Products-II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: properties
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

3 Remembering board Explain exterior grade, interior grade, single

3 Remembering board Explain exterior grade, interior grade, single is an official topic in Module 1 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering board Explain exterior grade, interior grade, single using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering board explain exterior grade, interior grade, single should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering board Explain exterior grade, interior grade, single means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-3 Remembering board Explain exterior grade, interior grade, single definition/meaning. steps, application. Technical terms English-.

layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B

layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B is an official topic in Module 1 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, layer, three layer, multi-layer, graded, 7 remembering extrusion pressed and flat presses p.b explain the schematic outline for p.b should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഒരു layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B ഏകദണ്ഡം definition/meaning ഏകദണ്ഡം. ഏകദണ്ഡം ഏകദണ്ഡം ഏകദണ്ഡം, steps, application ഏകദണ്ഡം ഏകദണ്ഡം ഏകദണ്ഡം. Technical terms English-ഏകദണ്ഡം ഏകദണ്ഡം.

3 Understanding manufacture

3 Understanding manufacture is an official topic in Module 1 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding manufacture using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding manufacture should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding manufacture means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding manufacture ഏകീകരണത്തിനുള്ള
definition/meaning നൽകുന്നു. ഏകീകരണത്തിന്റെ ഏകീകരണ, steps,
application നൽകുന്നു. Technical terms English-ൽ നൽകുന്നു
നൽകുന്നു.

Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer.

Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer. is an official topic in Module 1 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline need of drying of pb 2 applying block board types - block method, strip method. assembling of block board core. continuous gluing, spot gluing. decorative and non-decorative block board flush door types - solid core flush door, cellular core (cellulose) flush door. adhesive used. chipping and rebate cutting, facing testing and properties of block board & flush door properties of block board & flush door. alternative wetting and drying test, knife test, boiling test of blackboard-particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. classification of p.b - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed p.b uses of p.b, raw materials , adhesive - uf and pf resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, schematic out line p.b name factors. particle preparation, debarking cross cutting, splitting classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door

Aspect	What to prepare
	Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer, multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials, adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic outline P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer, multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials, adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic outline P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer.
 definition/meaning . steps, application . Technical terms English-
 .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

properties	Outline manufacturing process of block	
M1.01		3
Remembering	board	
	Explain exterior grade, interior grade, single layer, three layer, multi-layer, graded,	7
M1.02		
Remembering	extrusion pressed and flat presses P.B	
	Explain the schematic outline for P.B	
M1.03		3
Understanding	manufacture	
M1.04	Outline Need of Drying of PB	2
Applying		
Contents:		
Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door		
Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used.		
chipping and rebate cutting, facing Testing and properties of block board & flush door		
Properties of block board & flush door. Alternative wetting and drying test, knife test,		
boiling test of Blackboard-Particle board - introduction - history - particles,		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Disseminate pressing, finishing, testing and properties.

This module is presented from the official syllabus scope for Wood Products-II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Disseminate pressing, finishing, testing and properties.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip

Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip is an official topic in Module 2 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline chip storage and drying of chip 2 understanding summaries general considerations of chip should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Outline Chip storage and Drying of Chip 2
Understanding Summaries general considerations of chip ഏകദേശം അളവ്
definition/meaning എന്താണ് ചിപ്പ്. ചിപ്പ് ഉണ്ടാക്കുന്ന രീതി, steps, application
ചിപ്പ് ഉപയോഗിക്കുന്ന രീതി. Technical terms English- ചിപ്പ് ഉണ്ടാക്കുന്ന രീതി.

5 Applying blending

5 Applying blending is an official topic in Module 2 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying blending using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying blending should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Applying blending means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-5 Applying blending ഓരോന്നിനും നിർവ്വചനം/അർത്ഥം ഉണ്ടാകണം. ഓരോന്നിനും ഓരോന്നിനും, steps, application ഉണ്ടാകണം. Technical terms English-ൽ ഉണ്ടാകണം.

Explain particle board 2 Understanding Enumerate the properties and testing of

Explain particle board 2 Understanding Enumerate the properties and testing of is an official topic in Module 2 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain particle board 2 Understanding Enumerate the properties and testing of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain particle board 2 understanding enumerate the properties and testing of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain particle board 2 Understanding Enumerate the properties and testing of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നിരിക്കട്ടെ Explain particle board 2 Understanding Enumerate the properties and testing of പദാർത്ഥങ്ങളുടെ ഗുണങ്ങൾ definition/meaning പദാർത്ഥങ്ങളുടെ ഗുണങ്ങൾ. പദാർത്ഥങ്ങളുടെ ഗുണങ്ങൾ, steps, application പദാർത്ഥങ്ങളുടെ ഗുണങ്ങൾ. Technical terms English-എന്നിരിക്കട്ടെ പദാർത്ഥങ്ങളുടെ ഗുണങ്ങൾ.

5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation,

5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft

vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation, is an official topic in Module 2 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending

strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation, using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 remembering particle board storage of chips - horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. chip blending - definition, general consideration, batch operation, continuous operation, mat laying - terms and failures in volumetric and gravimetric dosing. dosing of resinated chips. volumetric and gravimetric dosing. dosing of resinated chips volumetric dosing and control of the weight of the chips. spreader heads and spreading machines, resultant mat- pre-pressing and pressing of p.b - pre pressing methods - advantages and its disadvantages. pressing - two wetting methods of mat, four particular proceedings of the hot platen press. three methods to achieve short pressing time in single day light press- effect of press platen temperature on physical properties of p.b. effect of curing time on physical properties of p.b, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of p.b. characteristics of uf resin with pf resin. list 3 caul less pressing process - continuous belt system. chip craft vertical extrusion process, lane wood horizontal extrusion processes calendar particleboard - mande - baisen system - p.b finishing - sizing sanding or thickening by wide belt sanding machine. physical and chemical changes during storage of p.b after leaving press. properties and testing of p.b - preparation and conditioning of test specimen, factors affecting the properties of p.b. physical properties, general properties, mechanical properties, technological properties and resistance against destruction. p.b testing methods - density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, modulus of elasticity, tensile strength perpendicular to the plane of board. state standard test result of p.b. explain the chipping of wood, pulping methods, mat formation, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of

Aspect	What to prepare
	resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen,

factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation,
 ഏകീകൃതമായ മാനദണ്ഡം definition/meaning
 ഏകീകൃതമായ മാനദണ്ഡം.
 steps, application
 ഏകീകൃതമായ മാനദണ്ഡം
 Technical terms English-
 ഏകീകൃതമായ മാനദണ്ഡം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Disseminate pressing, finishing, testing and properties.

M2.01	Outline Chip storage and Drying of Chip	2
Understanding		

	Summaries general considerations of chip	
M2.02		5
Applying		
	blending	

M2.03	Explain particle board	2
Understanding		

	Enumerate the properties and testing of	
M2.04		5
Remembering		

	particle board	
	Series Test - I	1

Contents:

Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting

methods, chip conveying methods. Chip blending - Definition, general consideration, batch

operation, continuous operation, Mat laying - Terms and failures in volumetric and

gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing

of resinated chips volumetric dosing and control of the weight of the chips.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

MODULE 3

manufacture of Insulation board and Hardboard

This module is presented from the official syllabus scope for Wood Products-II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: manufacture of Insulation board and Hardboard
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

6 Understanding manufacturing process of fiberboard

6 Understanding manufacturing process of fiberboard is an official topic in Module 3 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding manufacturing process of fiberboard using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding manufacturing process of fiberboard should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding manufacturing process of fiberboard means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-6 Understanding manufacturing process of fiberboard ഫൈബർബോർഡ് നിർമ്മാണം definition/meaning ഫൈബർബോർഡ് എന്താണ്. ഫൈബർബോർഡ് നിർമ്മാണ ഘട്ടങ്ങൾ, steps, application ഫൈബർബോർഡ് എവിടെ ഉപയോഗിക്കുന്നു. Technical terms English-ഫൈബർബോർഡ് നിർമ്മാണം.

Explain pulping processes

Explain pulping processes is an official topic in Module 3 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain pulping processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain pulping processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain pulping processes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain pulping processes തിരിച്ചറിയൽ നിർവ്വചനം/അർത്ഥം തിരിച്ചറിയൽ. തിരിച്ചറിയൽ തിരിച്ചറിയൽ, steps, application തിരിച്ചറിയൽ തിരിച്ചറിയൽ തിരിച്ചറിയൽ. Technical terms English- തിരിച്ചറിയൽ തിരിച്ചറിയൽ.

Describe mat formation

Describe mat formation is an official topic in Module 3 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe mat formation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe mat formation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe mat formation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
Describe mat formation
definition/meaning .
steps, application
Technical terms English-

Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and

Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and is an official topic in Module 3 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop testing of hardboard and mdf 3 applying fiber board insulation board mdf and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the m.c of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - fibreboard - block diagram of hardboard manufactures - sizing - mat formation and pressing - oil tempering, heat treatment, humidification - testing methods of fiber board. explain sandwich composition, bulking treatments and testing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and definition/meaning. Steps, application. Technical terms English-.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

manufacture of Insulation board and Hardboard

Recall Fiberboard, raw materials for fiberboard,

M3.01		6
Understanding		

manufacturing process of fiberboard

M3.02	Explain pulping processes	3
Applying		

M3.03	Describe mat formation	3
Applying		

M3.04	Develop testing of hardboard and MDF	3
Applying		

Contents:

Fiber board insulation board MDF and hard board - history and development of fiber board

- classifications of fiber board - modified wood raw materials - non wood fibrous raw

materials - properties and uses of different types of fiber board - properties of semi rigid

insulation board, rigid insulation board, hard boards - schematic outline of fiber board

manufacturer using wet, semi dry and dry process - chipping - magnetic separation and

screening of fibers - storage of chips and the importance of the M.C of chips during storage

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

properties of wood laminates.

This module is presented from the official syllabus scope for Wood Products-II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: properties of wood laminates.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Oriented strand board

Oriented strand board is an official topic in Module 4 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Oriented strand board using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, oriented strand board should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Oriented strand board means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓറിയന്റേഡ് സ്റ്റാൻഡ് ബോർഡ് ഓറിയന്റേഡ് സ്റ്റാൻഡ് ബോർഡ് definition/meaning ഓറിയന്റേഡ് സ്റ്റാൻഡ് ബോർഡ്. ഓറിയന്റേഡ് സ്റ്റാൻഡ് ബോർഡ്, steps, application ഓറിയന്റേഡ് സ്റ്റാൻഡ് ബോർഡ്. Technical terms English-ഓറിയന്റേഡ് സ്റ്റാൻഡ് ബോർഡ്.

Outline Densified wood

Outline Densified wood is an official topic in Module 4 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Densified wood using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline densified wood should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Densified wood means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓൗട്ലൈൻ ഡെൻസിഫൈഡ് വുഡ് ഏകദേശം ഏകദേശം
definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം, steps, application
ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-ഓൗട്ലൈൻ ഏകദേശം.

Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-Define sandwich Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: Sl. No Website Link

https://www.researchgate.net/publication/265168265_Timber_and_wood_products 1

http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3

http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4

https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5

<http://www.fao.org/3/an647e/an647e00.pdf>

Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-Define sandwich Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: Sl. No Website Link

https://www.researchgate.net/publication/265168265_Timber_and_wood_products 1
http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3
http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4
https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5
<http://www.fao.org/3/an647e/an647e00.pdf> is an official topic in Module 4 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-Define sand witch Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: SI. No Website Link
https://www.researchgate.net/publication/265168265_Timber_and_wood_products 1
http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3
http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4
https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5
<http://www.fao.org/3/an647e/an647e00.pdf> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain sandwich composition 5 applying oriented strand board - manufacturing methods - wood wool board - method of manufacture of wood wool board - properties and uses - cement bonded particleboard - method of manufacture - moulded particle board - werzalit process - densified wood - method of preparation of raw materials - densification of wood by impregnation and compression - metalized wood - compreg - dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-define sandwich composition - explain the raw materials - edge banding of particle board - laminated pb - bulking treatment - bulking treatments - bulking treatments such as salt treatment, sugar treatment, poly ethylene glycol treatment, wax treatment, phenolic resin treatment - properties and uses of impreg - acetylated wood - testing and properties. text / reference: t/r book title/author franz. f.p kollmann, principles of wood science and technology vol. ii - t1 springer thomas m malon, modern particle board and drying process of fiber board t2 manufacturing online resources: sl. no website link

https://www.researchgate.net/publication/265168265_timber_and_wood_produ 1 cts

http://www.kstr.lth.se/fileadmin/kstr/pdf_files/timber_engineering_2017/timber_and_engineered_wood_products.pdf 3

http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4

https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5

<http://www.fao.org/3/an647e/an647e00.pdf> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-Define sandwich Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: Sl. No Website Link https://www.researchgate.net/publication/265168265_Timber_and_wood_produ 1 cts http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3 http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4 https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5 http://www.fao.org/3/an647e/an647e00.pdf means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-Define sandwich Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: Sl. No Website Link
https://www.researchgate.net/publication/265168265_Timber_and_wood_products 1
http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3
http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4
https://www.fpl.fs.fed.us/documents/fplgtr/fpl_gtr190.pdf 5
<http://www.fao.org/3/an647e/an647e00.pdf>
definition/meaning . . . , steps, application Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

properties of wood laminates.

M4.01	Oriented strand board	5
Remembering		

M4.02	Outline Densified wood	4
Applying		

M4.03	Explain sandwich composition	5
Applying		

Series Test - II		1
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Contents:

Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded

Particleboard -

Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood -

method of Preparation of raw materials - densification of Wood by Impregnation and

Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction

in hygroscopicity, heat stabilisation & cross linking - Define sandwich

Composition -

Explain the raw materials - Edge banding of Particle Board - Laminated PB -

Bulking

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 3 Remembering board Explain exterior grade, interior grade, single. (3 marks)

Answer: Begin with the standard definition or identity of *3 Remembering board Explain exterior grade, interior grade, single*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B. (3 marks)

Answer: Begin with the standard definition or identity of *layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Understanding manufacture. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding manufacture.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive - UF and PF resin, additives - water

repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer, multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials, adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 5 Applying blending. (3 marks)

Answer: Begin with the standard definition or identity of *5 Applying blending*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain particle board 2 Understanding Enumerate the properties and testing of. (3 marks)

Answer: Begin with the standard definition or identity of *Explain particle board 2 Understanding Enumerate the properties and testing of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B

after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation,. (3 marks)

Answer: Begin with the standard definition or identity of 5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation,. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 6 Understanding manufacturing process of fiberboard. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding manufacturing process of fiberboard*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain pulping processes. (3 marks)

Answer: Begin with the standard definition or identity of *Explain pulping processes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Describe mat formation. (3 marks)

Answer: Begin with the standard definition or identity of *Describe mat formation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry

process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and. (3 marks)

Answer: Begin with the standard definition or identity of *Develop testing of hardboard and MDF* 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain Oriented strand board. (3 marks)

Answer: Begin with the standard definition or identity of *Oriented strand board*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Outline Densified wood. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Densified wood*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking - Define sandwich Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: SI. No Website Link

https://www.researchgate.net/publication/265168265_Timber_and_wood_products 1 cts

http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3

http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4

https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5

<http://www.fao.org/3/an647e/an647e00.pdf>. (3 marks)

1-mark definition: State the meaning of **3 Remembering board Explain exterior grade, interior grade, single.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **6 Understanding manufacturing process of fiberboard.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Oriented strand board.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link
- https://www.researchgate.net/publication/265168265_Timber_and_wood_produ
- http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf
- http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf
- https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf
<http://www.fao.org/3/an647e/an647e00.pdf>

IN-PAGE SEARCH

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